

Yuan (元), Primitive Distinction, and the Operational Descriptive Horizon: A Frozen Coarse-to-Fine Research Chain

“元”（暂名）、原始区分与操作描述视界：一条由粗到细的冻结研究链

Youfu Wang (王有福)

Independent Researcher, OCN Research

ORCID: 0009-0005-5998-6024

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Abstract

This structural note freezes the current lower boundary reached in a continuing coarse-to-fine research program on observation, distinguishability, probability, admissible state spaces, dynamics, and finite readout. A previous reduction identified primitive distinction as the least non-trivial operational structure that remains after probability, event algebra, observer identity, and readout labels are progressively removed. The present revision sharpens that result by refusing to identify the pre-distinction side of the boundary with a “state,” an “object,” a “field,” or an intrinsic “capacity for distinction.” Instead, the provisional Chinese term **Yuan (元)** is introduced solely as a low-commitment placeholder for whatever, if anything, lies on the not-yet-described side of the first expressible distinction. Yuan is not proposed as a physical entity and its existence is not asserted. Its function is to mark an unresolved position in the research map.

The first layer that can presently be formalized without immediately re-importing hidden assumptions is a **relational distinction**: a non-trivial test, comparison, or interaction produces different operational classes. From such a distinction one obtains a primitive cut, the minimal Boolean event algebra, probability assignments, observer/test families, refinement of observational resolution, disclosure of latent substructure, relational/admissible state organization, dynamics, projection/coarse-graining, and finite readout. The paper therefore relocates the **operational descriptive horizon** to the interface between Yuan and the first mathematically expressible relational distinction. Below that horizon, assigning a property to Yuan already introduces a distinction; above it, distinctions can be represented and tested. The horizon is a limit of the present descriptive framework, not a proof that reality ends there.

The paper also freezes the current full research chain in simple form. Quantum coherent superposition remains **PARKED / UNMAPPED** rather than being forced into this hierarchy, and an observer-independent unique-actual-state hypothesis remains **OPEN / UNPROVEN**. The next scientific task is no longer unconstrained descent. It is to test whether increasing observer resolution can split a coarse equivalence class into substructures with independently measurable responses. If such a split is found, the descriptive horizon moves; if not, the coarser operational description survives its death gate.

中文摘要

本文用于固化当前“由粗到细、再向更深层约化”研究所达到的暂时边界。此前的推理把概率、事件代数、观察者实体和 0/1 标签逐层删除后，发现“原始区分”是仍保留非平凡操作内容的最低可数学化结构。本次修正进一步强调：不能把区分之前的未知层直接命名为“状态”“对象”“场”或“可区分能力”，因为这些词都会提前加入尚未证明的性质，并立即引出“这种能力属于谁、来自哪里”的新假设。

因此，本文暂时采用中文技术占位词“元”。这里的“元”只表示：如果第一次可表达区分之前仍存在某种更深结构，当前理论暂以“元”标记那个未知位置。“元”不被预设为状态、对象、关系、观察者、能力、物质、信息或任何已知物理实体；本文也不宣称“元”一定存在。真正第一个可以在现有语言中被数学化的层级，是 **关系性**：某种测试、比较、相互作用或参照使原本属于同一操作等价类的候选产生不同响应。

在此基础上可以依次建立：原始切分 → 事件代数 → 观察/测试族 → 概率包络 → 观察分辨率细化 → 隐态显化 → 关系/允许态结构 → 动力学/流 → 投影/粗粒化 → 0/1、8、12 等有限读出。本文把 操作描述 视界更准确地放在“元”与“第一次可表达关系性区分”之间：视界以下，当前语言一旦试图给“元”赋予性质，就已经重新引入区分；视界以上，关系、分割、事件、概率和状态开始能够被表示和检验。这个视界只是当前描述能力的边界，不等于存在的边界。

量子相干叠加继续保持 **PARKED / UNMAPPED**（暂缓 / 未映射），不强行放入这条层级链；观察者无关的唯一实际态假说继续保持 **OPEN / UNPROVEN**（开放 / 未证实）。下一阶段应从向下猜测转向向上验证：通过增加新的观察维度或干预，检验一个粗粒等价类能否裂分出具有独立可测响应的细结构。

Keywords: Yuan (provisional term); primitive distinction; relational distinction; operational descriptive horizon; observer resolution; event algebra; operational equivalence; coarse-graining; latent-state disclosure; probability envelope; foundations of physics; operational models

1. Introduction

A foundational research program can stop too early and mistake an effective description for a final ontology, or it can descend indefinitely by naming lower layers without adding mathematical necessity or operational content. The present note is intended to avoid both errors.

Two preceding papers froze successive levels of the current research chain. The first separated a pre-projection admissible-state layer from downstream finite readouts [1]. The second formalized observer-relative undifferentiated probability envelopes and retained a deeper observer-independent actual-state completion as an open, empirically underdetermined possibility [2]. Subsequent reasoning then showed that probability presupposes a measurable event structure, that a minimal event algebra can be generated by a primitive cut, and that an observer can be de-personalized into a family of accessible tests or discriminators.

A further conceptual problem then appeared. Calling the deepest remaining layer a “capacity for distinction” already attributes a capacity to something and immediately raises an ungrounded ownership question: is the capacity intrinsic to the candidate object, supplied by an observer, or generated by their relation? Likewise, calling the pre-distinction side a “state” assumes a state space before one has been derived.

To avoid that premature commitment, the present paper introduces the provisional term **Yuan (元)** as a placeholder only. The central question becomes:

What is the first structure that can be expressed without presupposing the very distinctions the analysis is trying to derive?

Our current answer is not “Yuan itself.” Yuan remains undefined. The first formalizable layer is a **relational distinction**: some test, comparison, or interaction yields at least two operationally different responses. This defines the present operational descriptive horizon.

2. Scope and explicit non-claims

This document is a structural preprint and research-chain checkpoint. It does **not** claim:

that Yuan (元) is the ultimate ontology of nature;

that Yuan is a state, object, field, information substrate, observer, rule, or physical substance;

that Yuan is known to exist;

that consciousness is required for physical reality;

that binary 0/1 is fundamental;

that every refined label corresponds to a new independent physical state;

that quantum coherent superposition is false;

that a deeper observer-independent actual state has been established;

that the present framework evades Bell, contextuality, or other quantum-foundational constraints.

The strong claim is narrower: **the present descriptive framework first acquires non-trivial operational content when a non-trivial distinction becomes expressible**. Any statement about what lies before that first distinction must be treated as underdetermined unless a new formalism or experiment supplies additional structure.

3. Yuan (元) as a provisional pre-distinction placeholder

We use the symbol

$$\Xi_0 \equiv \text{Yuan (元)} \quad (1)$$

only as a placeholder.

Definition 1 (Yuan placeholder). Yuan denotes the unresolved position in the research map that lies prior to the first currently expressible non-trivial distinction. No intrinsic property is assigned to it.

Accordingly, the following statements are deliberately **not** assumed:

$$\Xi_0 \in \mathcal{S}$$

because this would already assume a state space; nor do we assume that Yuan possesses a discriminator, a probability distribution, a metric, a topology, a time variable, or an observer-independent identity.

The name is therefore semantic scaffolding, not ontology. If a later theory derives a better structure, Yuan should be replaced or refined rather than defended as a fixed substance.

3.1 Why “no distinction” is not the same as “before distinction”

Once a predicate (D) has been defined, the pair

$$D \mid \neg D \quad (2)$$

already forms a distinction. Therefore the statement “there is no distinction” belongs to a language in which the distinction predicate has already been introduced. The pre-distinction side of the present horizon is more accurately described as **not-yet-defined with respect to that predicate**, not as the false value of the predicate.

Thus:

$$\text{undefined with respect to } D \neq \neg D \quad (3)$$

This is a meta-descriptive observation, not a new physical law.

4. First formalizable layer: relational distinction

Because no intrinsic “capacity” is assigned to Yuan, distinguishability is treated first as a relational property.

Let Ω be a candidate domain at the first level where a domain representation has become meaningful. Let a test or comparison relation be represented by

$$d: \Omega \rightarrow Y_d \quad (4)$$

A non-trivial operational distinction exists when

$$|\text{im } d| \geq 2 \quad (5)$$

The key point is that we do not say that Ω intrinsically “has distinguishability,” nor that an observer creates reality. We say only that within a declared relation/test structure, different operational responses can occur.

An observer/interface O can then be represented by a family of accessible discriminators

$$\mathcal{D}_O = \{d_1, d_2, \dots\} \quad (6)$$

No conscious agent is required. A detector, reference system, intervention set, or comparison protocol can play the role of O .

Definition 2 (observer-induced equivalence). For (x, x') ,

$$x \sim_O x' \Leftrightarrow d(x) = d(x') \text{ for every } d \in \mathcal{D}_O \quad (7)$$

The effective state space available to O is the quotient

$$\Omega_O = \Omega / \sim_O \quad (8)$$

Proposition 1. Non-trivial empirical content requires a non-constant discriminator

If every $d \in \mathcal{D}_O$ is constant on Ω , then Ω_O has exactly one equivalence class. Conversely, if Ω_O contains at least two equivalence classes, at least one accessible discriminator is non-constant.

This elementary result defines the first formalizable operational layer. It does not define Yuan.

5. Primitive cut and the first event algebra

Take a non-constant binary discriminator

$$d: \Omega \rightarrow \{L, R\} \quad (9)$$

Define

$$A = d^{-1}(L), \quad A^c = d^{-1}(R) \quad (10)$$

The pair

$$\{A, A^c\} \quad (11)$$

is a primitive cut. Closing it under the usual Boolean operations generates

$$\Sigma_A = \{\emptyset, A, A^c, \Omega\} \quad (12)$$

Proposition 2. Minimal non-trivial event algebra

The Boolean algebra generated by one non-trivial binary distinction contains exactly four elements: $\emptyset, A, A^c, \Omega$.

The “two” generated here is a minimal two-class distinction, not a claim that physical ontology is binary. We therefore keep separate

$$2_{\text{distinction}} \neq 2_{\text{readout}} \quad (13)$$

6. Event structure before probability

A Kolmogorov probability space is a triple

$$(\Omega, \Sigma, \mu) \quad (14)$$

where Σ is a σ -algebra and μ is a probability measure defined on Σ [3]. Within that mathematical model, probability therefore presupposes an event domain.

At the coarsest event structure,

$$\Sigma_0 = \{\emptyset, \Omega\} \quad (15)$$

only

$$\mu(\emptyset) = 0, \quad \mu(\Omega) = 1 \quad (16)$$

are available. A non-trivial probability vector appears only after non-trivial alternatives have become measurable.

Thus, above the descriptive horizon, the current order is

$$\text{relational distinction} \rightarrow \text{event structure} \rightarrow \text{probability} \quad (17)$$

This is standard mathematics used here to order the research chain correctly.

7. Observer resolution and latent-state disclosure

Suppose

$$\mathcal{D}_{O_1} \subseteq \mathcal{D}_{O_2} \quad (18)$$

Then O_2 has at least as much distinguishing power as O_1 .

Proposition 3. Refinement monotonicity

If $\mathcal{D}_{O_1} \subseteq \mathcal{D}_{O_2}$, then

$$x \sim_{O_2} x' \Rightarrow x \sim_{O_1} x' \quad (19)$$

Thus a coarse observer can see only $\{0, 1\}$, while a refined observer may split the same coarse classes into $\{0A, 0B, 1A, 1B\}$, and still finer interfaces may continue to split them.

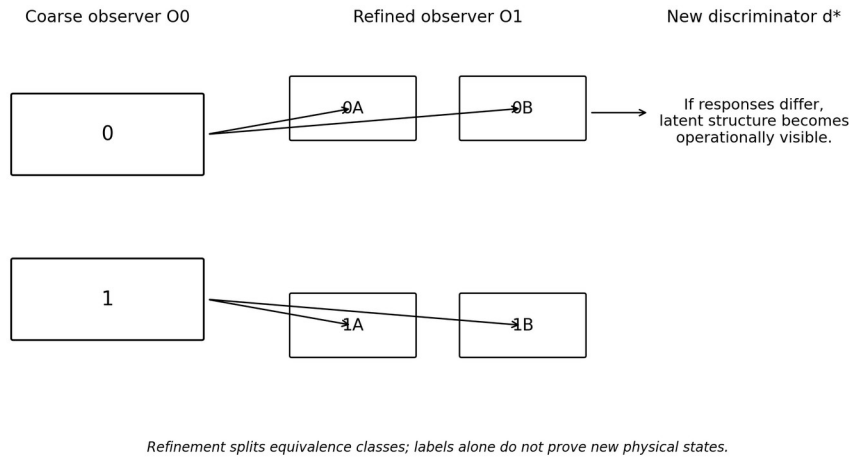


Figure 2. Observer-resolution refinement splits coarse equivalence classes.

Definition 3 (latent relative to an interface). Two candidates x, x' are latent-distinct relative to O if $x \sim_O x'$, but there exists an expanded interface O' containing an additional discriminator d^* for which

$$d^*(x) \neq d^*(x') \quad (20)$$

This distinguishes mere representational relabeling from operational latent-state disclosure. Refined labels acquire independent physical significance only when a reproducible intervention produces different observable responses.

8. Coarse, refined, and ontic descriptions

The previous A/B language is now separated into three roles:

A_{coarse} : a coarse operational probability/equivalence description;

B_{refined} : a finer operational description produced by richer observation or intervention;

B_{ontic} : the separate hypothesis that an observer-independent actual state exists beneath all operational descriptions.

The first two may be related by coarse-graining:

$$B_{\text{refined}} \rightarrow A_{\text{coarse}} \quad (21)$$

They need not be competing theories. A coarse observer and a fine observer can validly assign different levels of certainty or different effective state counts to the same process.

By contrast,

$$B_{\text{ontic}} = \text{OPEN} / \text{UNPROVEN} \quad (22)$$

remains outside the established chain. It requires an independently testable consequence before it can be promoted from ontological possibility to physics.

9. The operational descriptive horizon

We now relocate the descriptive horizon more carefully.

Definition 4 (operational descriptive horizon). The operational descriptive horizon is the current boundary between (i) an unresolved placeholder to which no non-trivial property can be assigned without reintroducing a distinction and (ii) the first relational structure in which a non-trivial distinction becomes expressible.

In the present notation:

$$\Xi_0(\text{Yuan, unresolved}) \mid \text{Operational Descriptive Horizon} \mid \text{first relational distinction} \quad (23)$$

This definition is intentionally asymmetric. We do **not** claim to have a mathematical model of Yuan. We only model the first structure above the horizon.

The horizon therefore means:

$$\text{limit of present descriptive resolution} \neq \text{limit of existence} \quad (24)$$

To say that Yuan “exists,” “does not exist,” “has a property,” or “lacks a property” already places it inside a language of distinctions. Accordingly, the scientifically disciplined position is underdetermination, not metaphysical affirmation or denial.

10. The frozen research chain

The current chain is summarized in Figure 1.

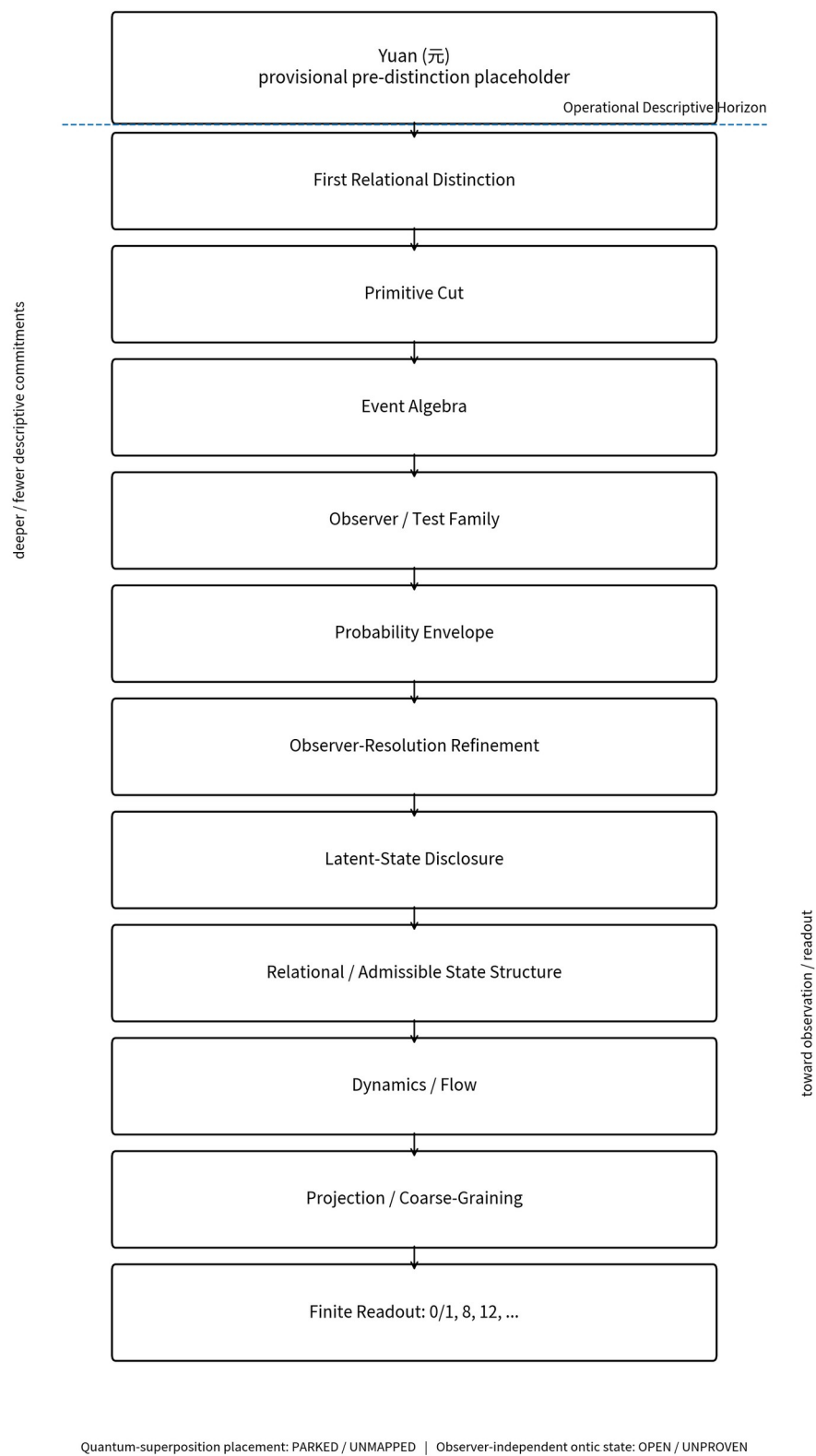
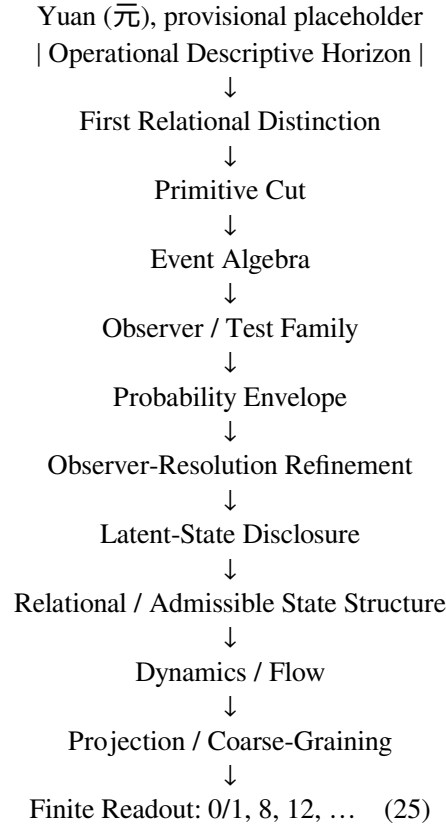


Figure 1. Current frozen research chain with Yuan as a provisional placeholder below the operational descriptive horizon.

In compact form:



10.1 One-sentence interpretation of the chain

Yuan (元): a placeholder for the presently undescribed side of the first expressible distinction; no ontology is assigned.

First relational distinction: a comparison, interaction, or test yields at least two operational responses.

Primitive cut: one non-trivial test divides a represented domain into two classes.

Event algebra: cuts are closed under logical operations to form measurable event structure.

Observer/test family: an interface is the collection of distinctions it can access.

Probability envelope: probabilities are assigned over measurable but incompletely resolved alternatives.

Observer-resolution refinement: additional tests generate finer operational partitions.

Latent-state disclosure: candidates previously aliased by a coarse interface become operationally distinguishable.

Relational/admissible structure: distinguishable classes acquire relations, coordinates, topology, or admissibility structure.

Dynamics/flow: allowed states acquire transformation rules and trajectories.

Projection/coarse-graining: richer descriptions are compressed into effective observational variables.

Finite readout: the interface returns a finite label set such as 0/1, 8, or 12.

This is a research map, not a derivation of nature. Each arrow remains independently defeasible.

11. Why quantum coherent superposition remains parked

A previous internal branch attempted to place quantum coherent superposition in the same coarse-to-fine hierarchy. That placement is deliberately suspended.

Quantum coherence carries complex-amplitude, phase, and interference structure that is not automatically ordered by the same classical refinement relation used above. Therefore:

$$C_{\text{quantum}} = \text{PARKED} / \text{UNMAPPED} \quad (26)$$

This means neither “false” nor “fundamental.” It means that forcing it into the present hierarchy would create a layer-mapping error before the relation is understood.

12. Relation to standard mathematics

The mathematical ingredients used above are standard and should not be overstated as new mathematics.

Kolmogorov probability places a measure on a σ -algebra of events [3]. Stone representation relates abstract Boolean algebras to set/topological representations [4]. Shannon information theory quantifies uncertainty over distinguishable alternatives [5]. Blackwell comparison provides a formal language for comparing statistical experiments by informational refinement [6]. Operational approaches in quantum foundations similarly emphasize preparations, transformations, measurements, and operational equivalence before ontological interpretation [7,8].

The present contribution is the **research-chain synthesis and boundary discipline**: the explicit separation of Yuan as an unresolved placeholder from the first mathematically expressible relational distinction, followed by a frozen chain through event structure, probability, observer refinement, latent-state tests, dynamics, and readout.

13. Validation program after the boundary

Because further unconstrained descent would largely become semantic speculation, the next priority should be upward validation.

Gate V1 - Minimal refinement

Choose a coarse map

$$q_0: \Lambda \rightarrow \{0, 1\} \quad (27)$$

and propose the smallest added discriminator d^* that splits one fiber of q_0 .

Gate V2 - Operational reality

For candidates λ, λ' in the same coarse fiber, search for an intervention u^* such that

$$P(y \mid \lambda, u^*) \neq P(y \mid \lambda', u^*) \quad (28)$$

Without Equation (28), the refined labels remain representational.

Gate V3 - Minimal observer resolution

Determine the smallest discriminator family needed to separate the target candidates. This converts “observe more carefully” into an optimization problem.

Gate V4 - Reconstruction sufficiency

If every relevant future response factors through the coarse state, prove the coarse description operationally sufficient for the declared task class. A negative result is valuable because it prevents artificial proliferation of hidden states.

Gate V5 - Reopening Yuan

The Yuan question should be reopened only if either:

a new formalism can characterize the pre-distinction side without circularly presupposing the distinction it aims to derive; or

an experiment reveals a new operational split that forces the descriptive horizon downward.

14. Discussion: what can and cannot be said about Yuan

Three positions must remain separate:

Nothing lies beyond the horizon. Not established.

A definite deeper entity lies beyond the horizon. Not established.

The present language cannot resolve what, if anything, lies beyond the horizon. This is the current position.

The provisional term Yuan is useful precisely because it avoids pretending to know which of the first two statements is correct. It marks an unknown slot in the research map. If later work finds that the slot should be represented by a relation, category, process, algebra, topology, physical state, or something else, the placeholder should be replaced.

This also clarifies why the boundary can feel like a loss of descriptive fidelity. Any attempt to characterize Yuan by saying “it is X” immediately introduces the distinction X/not-X. The act of description itself therefore moves the analysis back above the horizon. This is a property of the current descriptive scheme, not evidence for a mystical or ineffable physical substance.

15. Conclusion

The present work freezes a temporary lower boundary in a coarse-to-fine research chain. The earlier phrase “capacity for distinction” is withdrawn as a candidate deepest-layer name because it prematurely attributes a capacity whose source is unknown. The pre-distinction side is instead marked by the provisional placeholder **Yuan (元)**, with no claim that Yuan is a state, object, field, observer, rule, information substrate, or physical entity.

The first layer that can currently be formalized is relational distinction. From it one can construct a primitive cut, event algebra, probability, observer/test families, resolution refinement, latent-state disclosure, relational/admissible structure, dynamics, projection, and finite readout. The operational descriptive horizon lies between Yuan and that first expressible relational distinction.

The central boundary statement is therefore:

$$\text{descriptive horizon} \neq \text{ontological end} \quad (29)$$

The research program should now stop unconstrained descent and move upward toward validation. If a finer observer or intervention reveals independently measurable substructure inside a coarse equivalence class, the horizon moves and the chain deepens. If no such refinement survives testing, the existing coarse description remains sufficient. Yuan should be revisited only when new mathematics, new experimental leverage, or a genuinely new conceptual distinction makes the question better posed.

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Status statement

This document is a structural preprint / formal research note. It freezes a temporary descriptive boundary and a provisional term, Yuan (元), for the presently unresolved side of that boundary. It does not claim experimental validation of a final ontology and does not claim that no deeper reality exists.